

# Sho Sakaue

Salt Lake City, Utah | [sho.sakaue0808@gmail.com](mailto:sho.sakaue0808@gmail.com) | [LinkedIn](#)

## Education

### University of Utah – Bachelor of Science in Computer Science & Applied Math (December 2027)

- **Mathematics:** Linear Algebra, Numerical Analysis, Probability & Statistics
- **CS:** Machine Learning & Deep Learning, Data Structures & Algorithms, Model of Computing
- **GPA:** 3.95/4.0 Dean's List (all semesters)

## Skills

- **Programming:** Python (Pytorch), Java, C++, C#, SQL
- **Developer tools:** Git/GitHub, Docker
- **Software Engineering:** Object-Oriented Programming, MVC Architecture, Multithreaded Program
- **Language:** Japanese, English

## Work and Research Experiences

### Research Assistant — STARS Lab, University of Utah — UROP funded (August 2025 - Present)

- Built an automated alloy simulation workflow using ASE and Machine Learning Interatomic Potentials (MLIPs), integrating structure relaxation and post-processing for grain-boundary analysis.
- Reduced relaxation time per-configuration from DFT-scale hours to 3–5 minutes using MLIP-based simulations, significantly increasing evaluation throughput.
- Validated MLIP accuracy against DFT and classical potentials, achieving 0.1-order error in energy-based metrics (grain boundary, surface, and fracture energies).
- Performed MD-based tensile tests on various systems to generate separation distance vs. fracture energy curves, enabling quantitative fracture characterization.

## Projects

### Parameter-Efficient Fine-Tuning of Large Language Models (LoRA/QLoRA)

- Implemented LoRA and QLoRA from the original papers in PyTorch, manually building the low-rank adaptor and quantization pipeline rather than relying solely on high-level libraries.
- Achieved 99% parameter reduction compared to full fine-tuning while maintaining competitive model performance.
- Fine-tuned a llama-2-7B model and benchmarked parameter-efficient methods against base and fully fine-tuned models, evaluating accuracy-performance tradeoffs.
- Conducted ablation experiments varying LoRA rank and quantization bit-width, analyzing their impact on accuracy, memory efficiency, and training behavior.

### Multiplayer Networked Game & HTTP Server — Software Practice (November 2025 - December 2025)

- Developed a distributed, multiplayer game client using C# and Blazor following the MVC architectural pattern.
- Made a multi-threaded TCP network controller to handle real-time JSON game state updates.
- Managed thread safety and concurrency using locks to prevent race conditions during asynchronous data processing.
- Integrated a MySQL database to store game history and player high scores.

## Competition / Hackathon

### MIT iQuHACK — Quantum Computer Hackathon (January 31, 2026 - February 1, 2026)

- Analyzed large quantum circuits (50–60 qubits) from QASM files and implemented quantum circuit simulations using Qiskit to identify hidden peak bitstrings and achieved 24th place in the BlueQubit challenge.
- Optimized performance through gate reordering and cancellation to lower circuit complexity and simulation runtime for 10,000 shots by about 30%.
- Designed statistical post-processing techniques to extract peak signal from near-uniform output distributions, including bitstring filtering by Hamming weight and bit-position frequency patterns, and achieved accurate inference with just 10,000 shots instead of exponential brute-force search.
- Developed candidate-selection heuristics by comparing Hamming distances across sampled outputs to rank likely peak solutions.